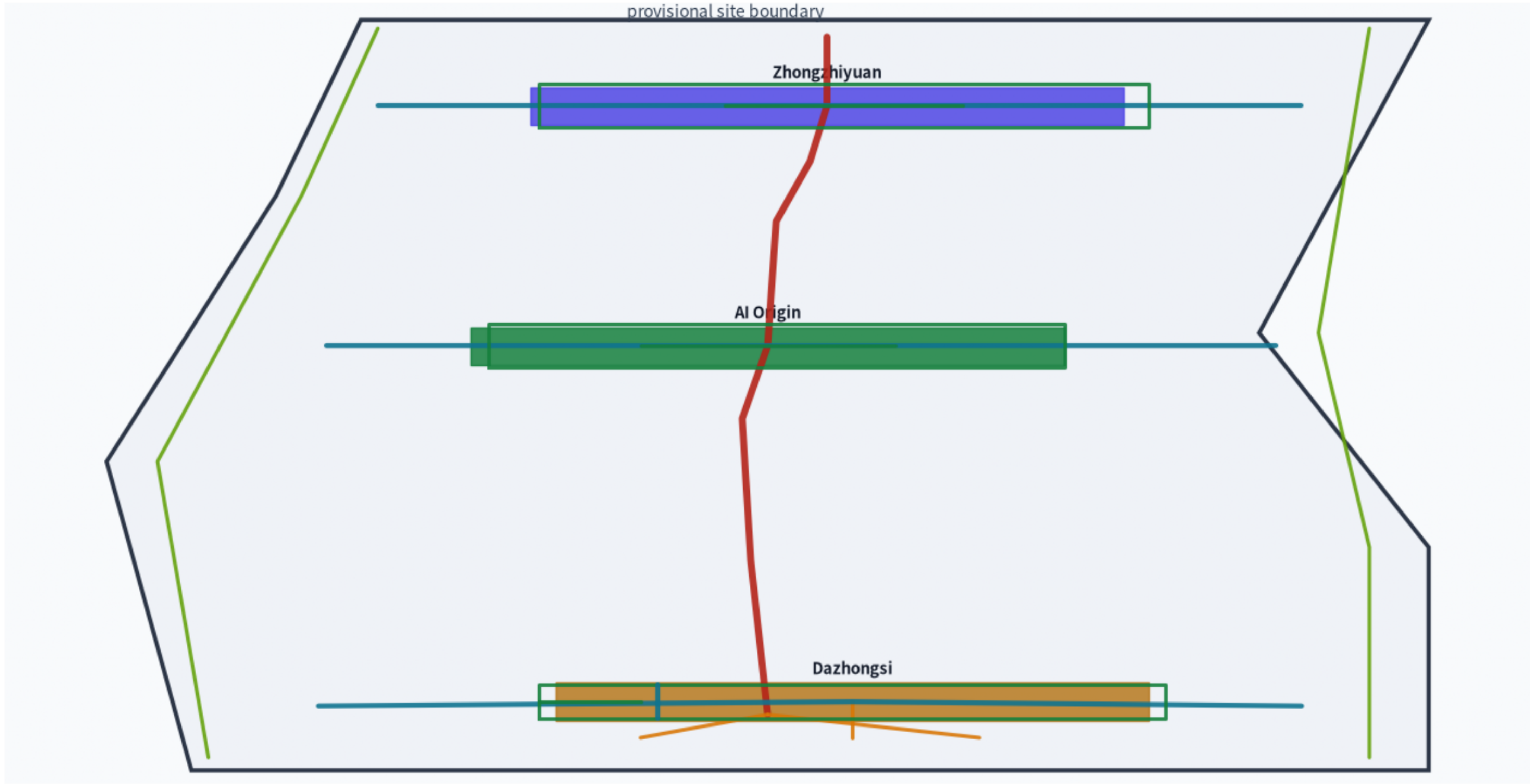


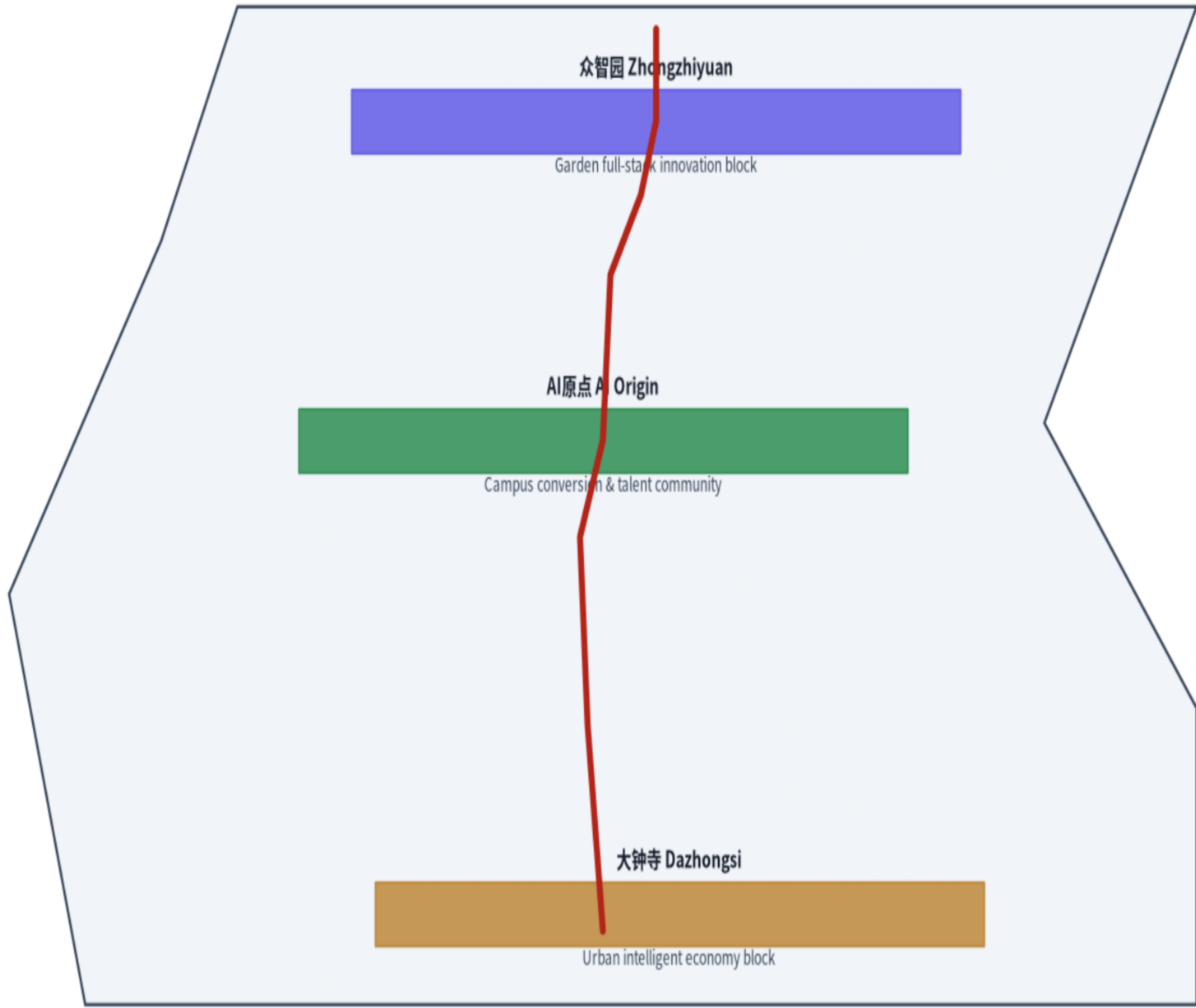
Site overview & conceptual network

Evidence chain and package



Three key areas + Physarum spine

Key areas and Physarum spine

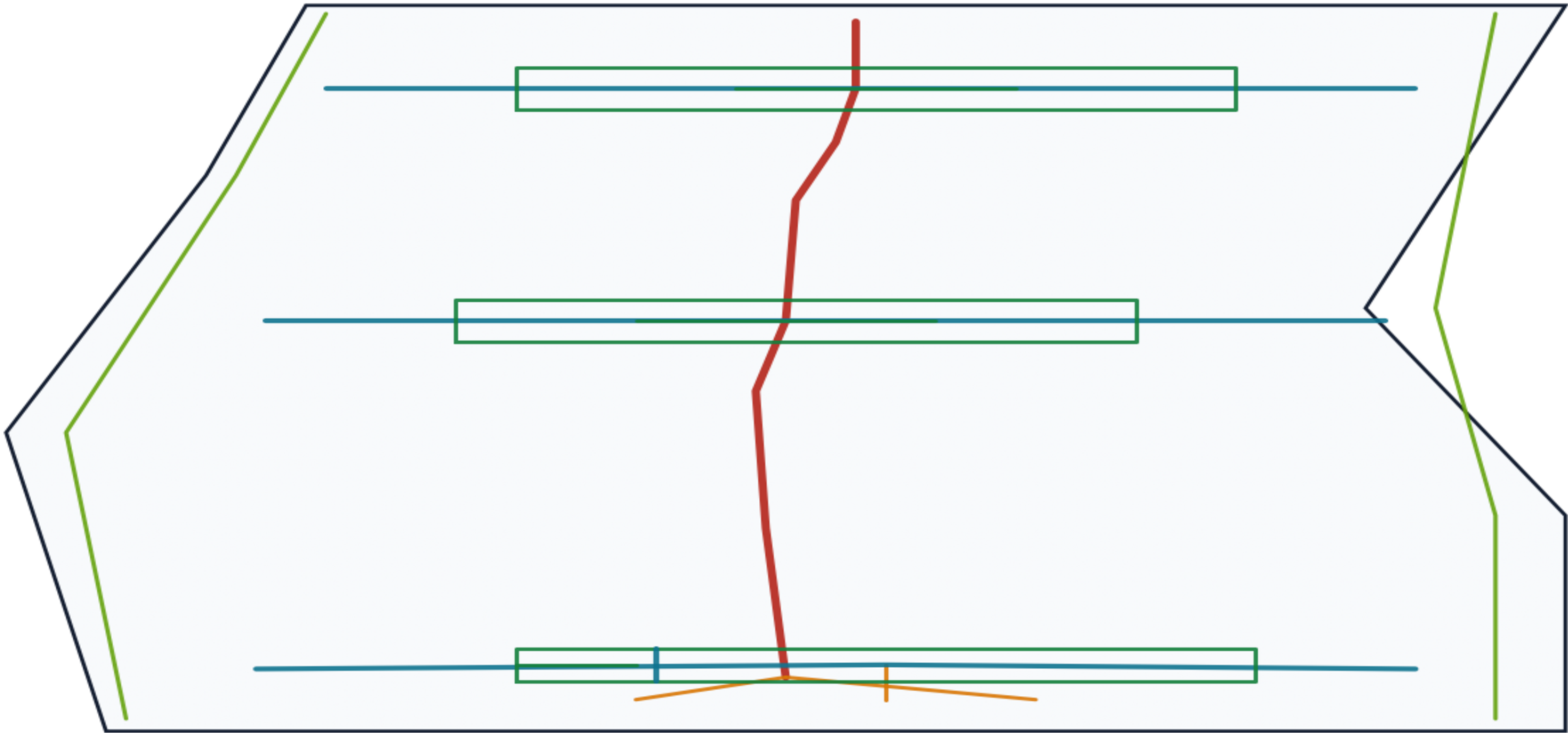


Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective | Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)

Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

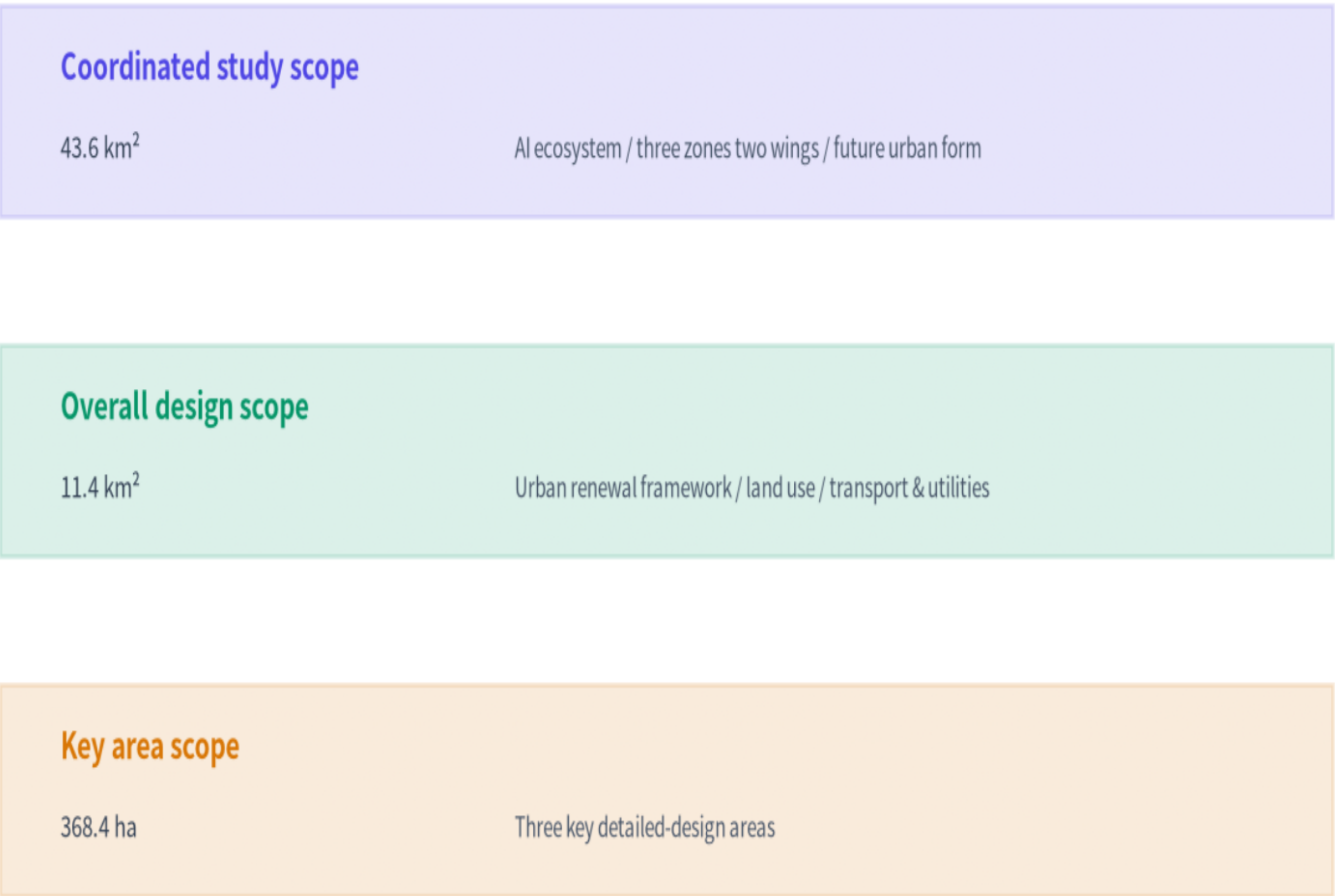
Mobility & blue-green public space

Mobility and blue-green public space



Three-level scope & spatial structure

Three-level scope and spatial structure



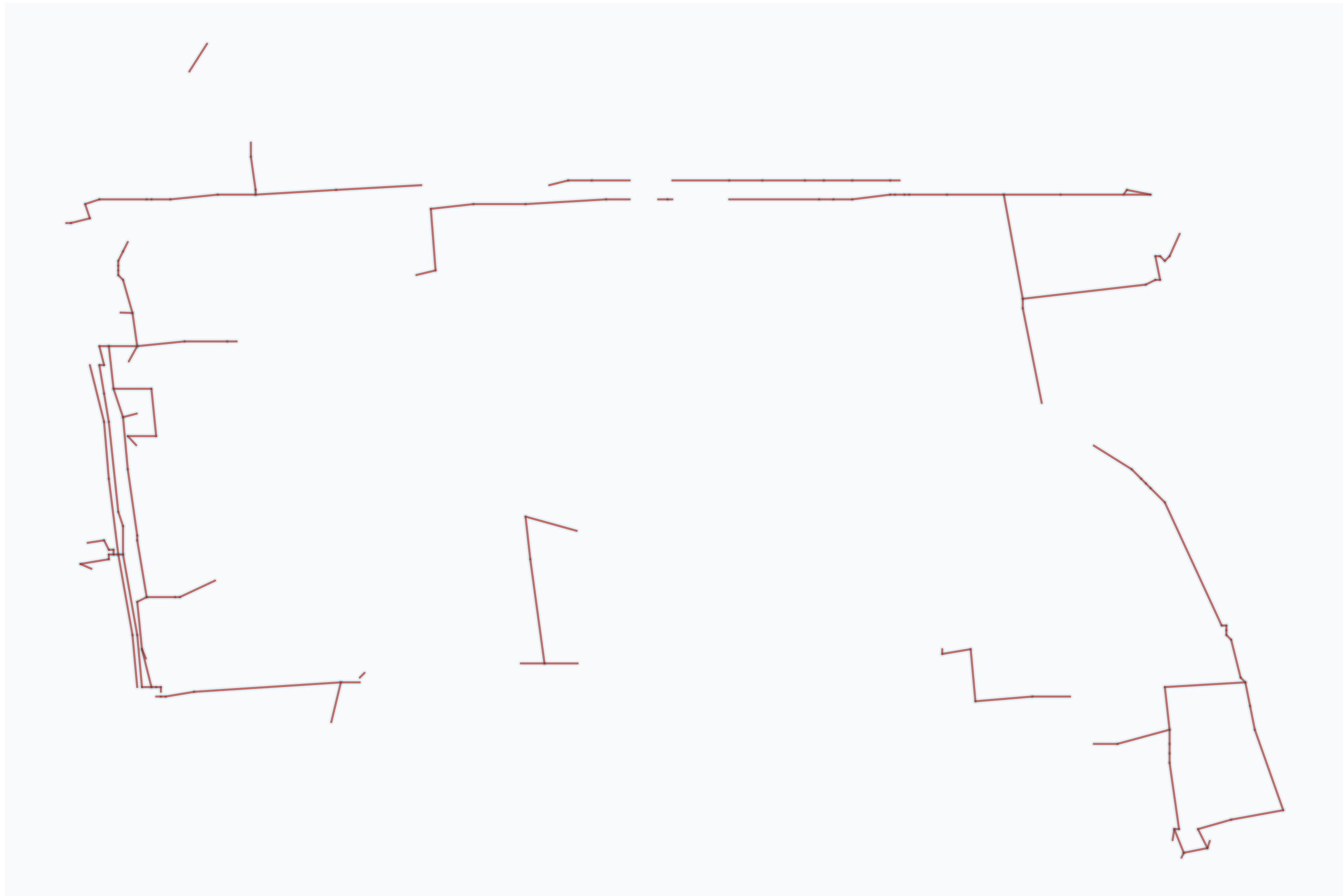
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

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Method & Results: Bio-Physarum Adaptive Network + NSGA-II

Technical route (Phase 0 → deliverables)

Real Physarum 167-edge skeleton (method validation)



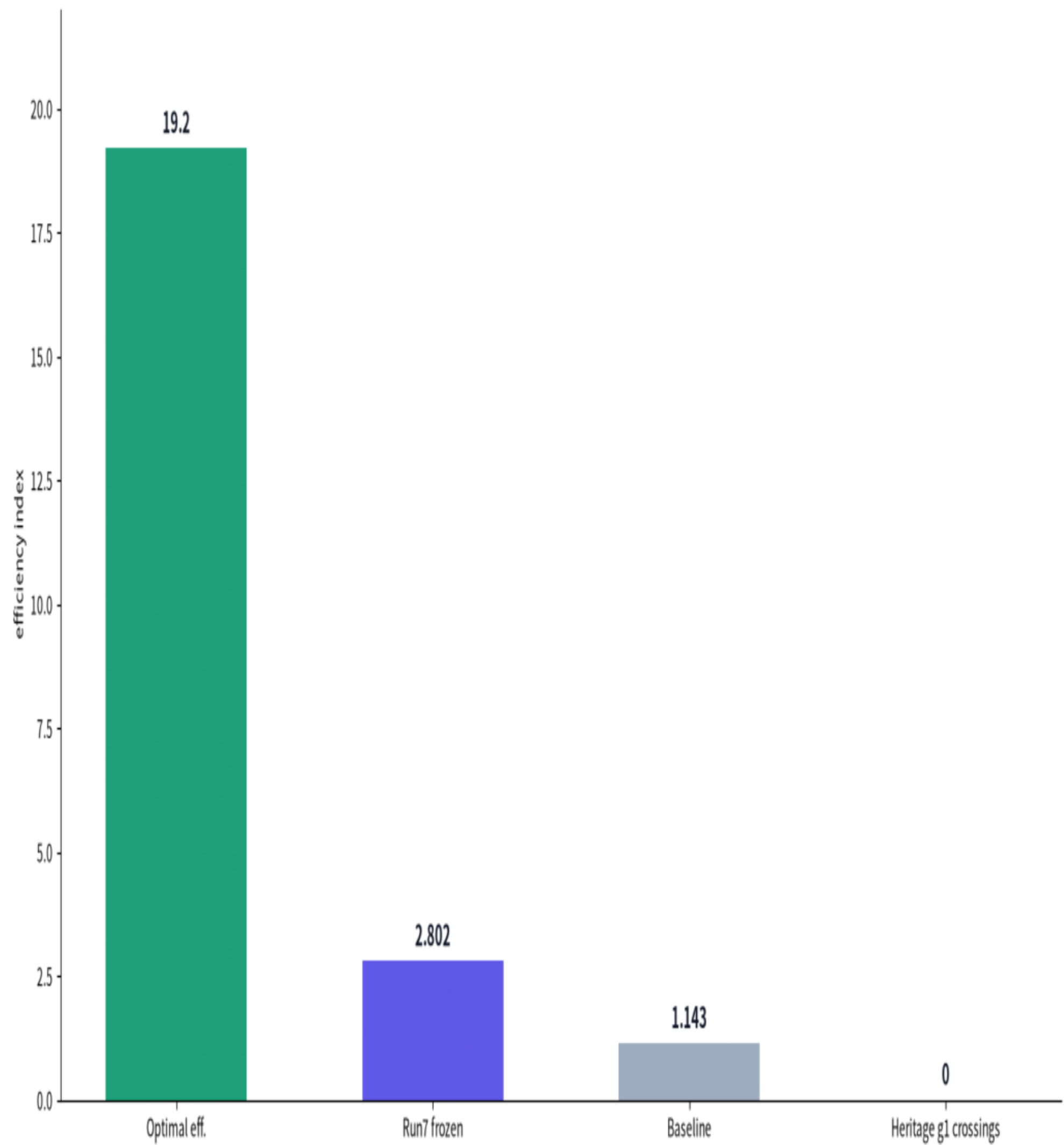
★ 黏面效率 1.143 → 19.20 (约 16.8×)

★ 主方案 Plan03 城市融合 UDS 80.34



Core metrics & method validation

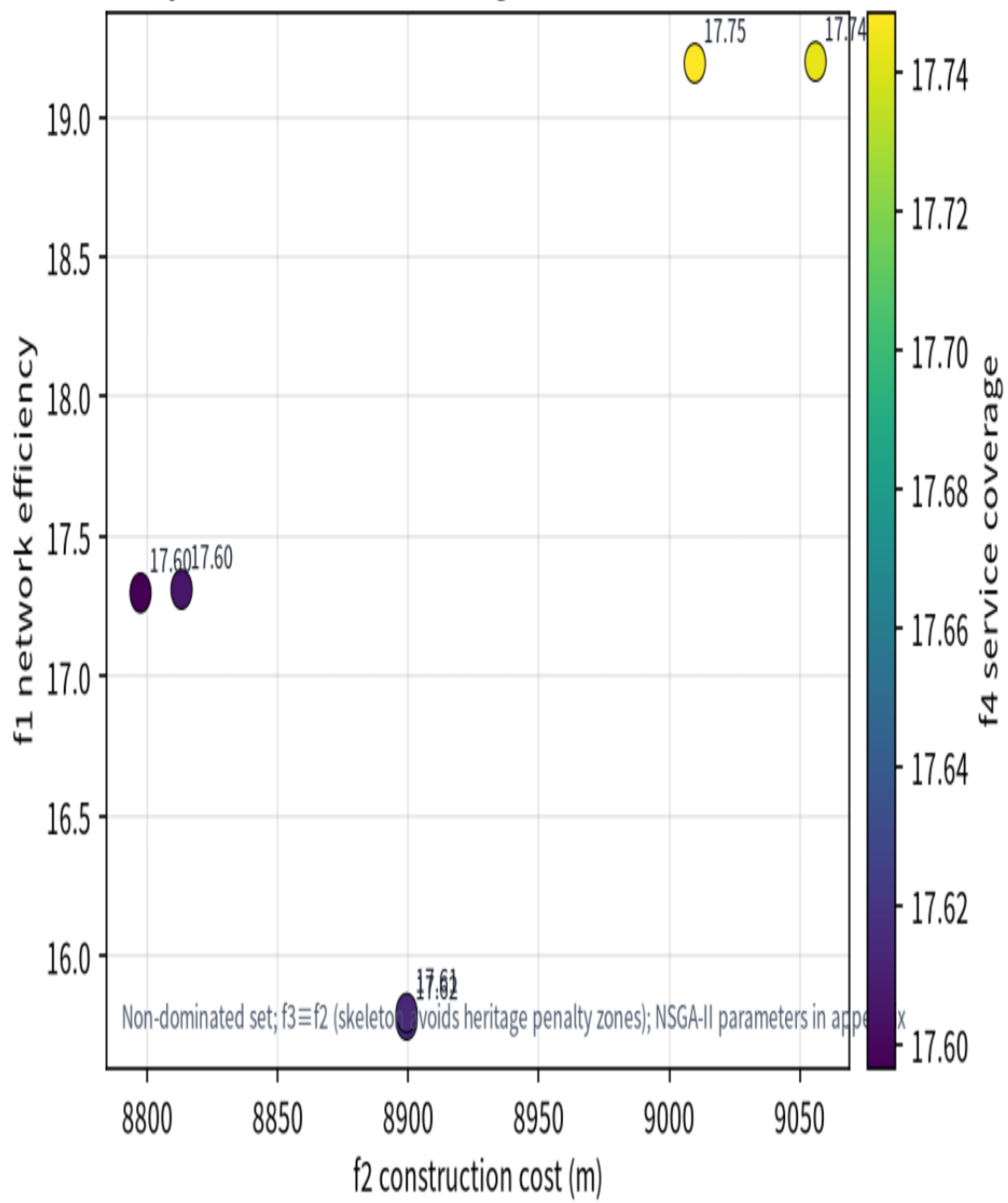
Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective U₃≡f2 degenerates to cost — "0 heritage crossings" is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective U₃≡f2 (degenerates to cost; no independent site heritage conclusion)

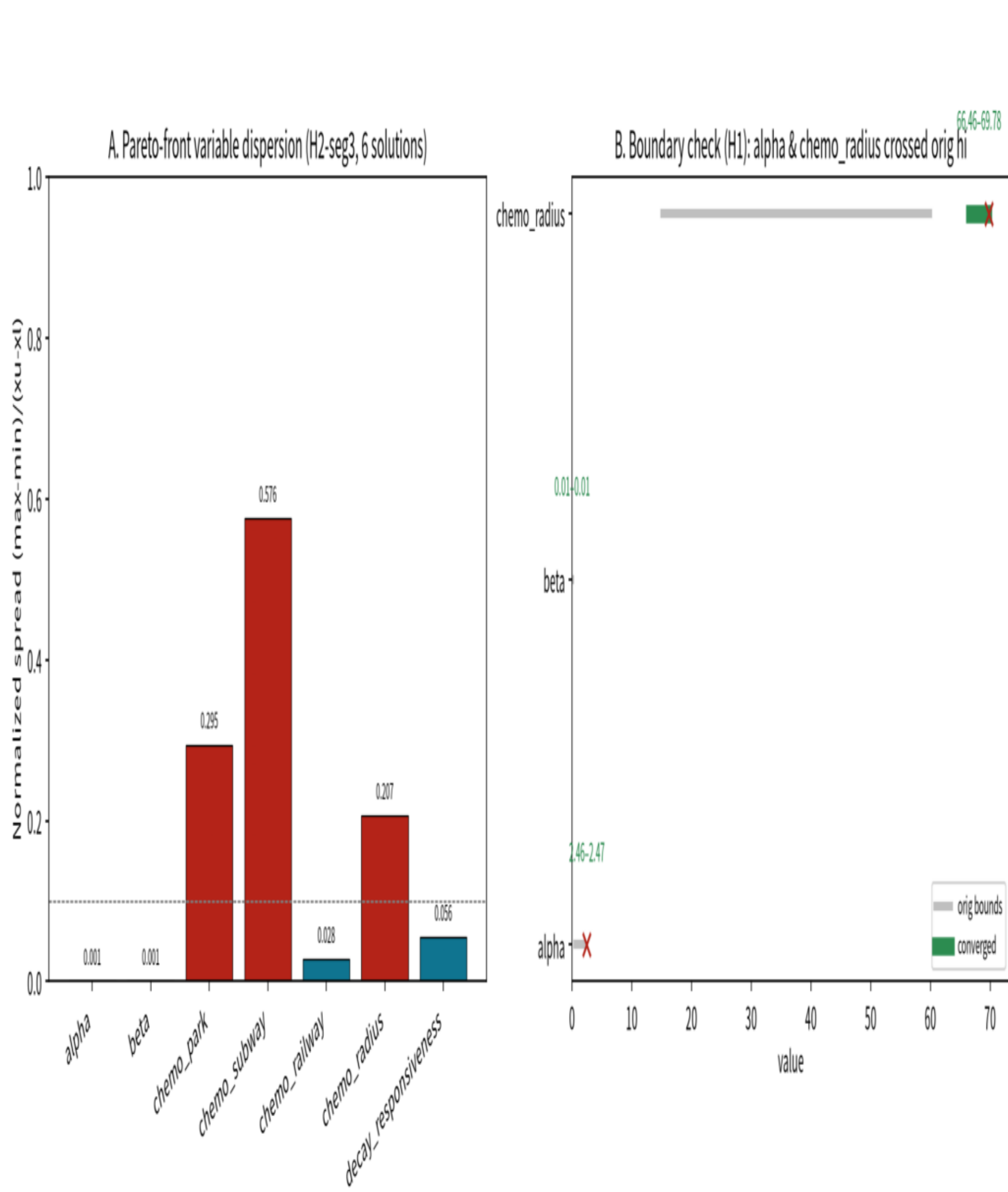
Four-objective Pareto front (6 sol.)

Four-objective Pareto front (H2-seg3, 6 non-dominated solutions)



Parameter sensitivity (suggestive)

Parameter sensitivity – Bio-Physarum + NSGA-II (real runs, method validation)



Suggestive phasing gantt (6 projects × 3 phases)

Suggestive phasing gantt (6 projects x 3 phases)



建议性分期, 无官方时间表 / suggestive phasing, no official schedule

19.20

Optimal eff.

1.143

Baseline

2.802

Run7 frozen

167

Skeleton edges

8813 m

Skeleton length

80.34

Plan03 UDS

69.1%

Permeable

25.2%

Green pen.